Nicolas Moreno Martinez

Math Student of Konrad Lorenz University

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Academic Background

Semester number eight / In my 4rd year mathematics major (2026-1) at Fundación Universitaria Konrad Lorenz, Bogotá Colombia, Interested in research.

Academic research

I am part of a research group led by Ph.D. José Alberto Vélez Marulanda, focused on Topological Data Analysis (TDA), where I have been working for one year. Our research explores applications of TDA methods on real datasets from Colombia. In my current project, we began studying flood risks in Colombia using a geometric big data framework. Beyond extensive data cleaning, we have implemented TDA techniques to analyze potential future flooded areas, regions at risk due to their topology, and possible relationships between zones. From the mathematical and topological perspective, we have worked with persistence diagrams, metrics, homology dimensions, Euler curves, and persistence landscapes. On the computational side, we employ parallel programming, vectorization, and comparative approaches to improve compilation time, using code partly in Python, with initial implementations in R, and future extensions planned in C++.

A portion of the code is available on my GitHub to promote open access, although I reserve the fundamental components for my research group.

I am currently working as a research assistant under Ph.D. José Alberto Vélez Marulanda in Topological Data Analysis. This work is directly connected to my undergraduate thesis, which focuses on applications of TDA combined with machine learning in financial domains.

Some parts of this project are also published on my GitHub to encourage free access to code, while I keep the core contributions reserved for academic purposes.

I developed a project for a psychology student that consisted of adapting a free artificial intelligence system to the point of challenging its morality, ethics, and human-like constraints, in order to debate sensitive topics such as abortion and gender. This project was mainly based on the architecture of artificial intelligence systems, modifying their APIs and then measuring the effectiveness of those changes.

I completed a summer program at the Delfín organization, during which I developed a research article on clustering, focusing specifically on two algorithms: K-means and DBSCAN. These algorithms belong to unsupervised machine learning techniques. In the article, I provide a detailed explanation and in-depth exploration of these methods, and I apply them to a real-world dataset. The data were extracted from ICFES, the institution in Colombia responsible for administering important national standardized tests, specifically the 'Saber 11' examination.

After cleaning the data in this project, I developed time series to apply clustering methods and analyze their differences. The time series comprises 31 semesters of available test results, from 2010-1 to 2025-2, excluding the first semester of 2016, and is expected to be updated to include the first semester of 2026. The goal of this article is to interpret the results in order to identify the most effective clustering approach. This analysis was conducted across twenty-five departments nationwide, producing promising results that align with the theoretical framework of educational levels in Colombia.

If there is some interest in my research repository `https://github.com/nikomm13`

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Areas of Expertise

- Advanced Mathematics: strong background in linear algebra, topology, statistics (regression methods, probabilistic models), and data analysis.
- Topological Data Analysis (TDA): persistence diagrams, persistent homology, Euler characteristic curves, persistence landscapes, and metric-based approaches.
- Programming and Computational Skills: proficient in Python and R, with experience in C++; parallel programming, vectorization, time complexity optimization, and algorithmic efficiency.
- Machine Learning and Data Science: application of supervised and unsupervised learning methods, integration of TDA with ML models, and analysis of large-scale datasets.
- Research and Collaboration: strong investigative skills, teamwork experience in academic projects, and interdisciplinary applications of mathematics and computer science.
- Open Science and Code Sharing: contributor to open-source projects via GitHub, balancing free access with preservation of core research contributions.

Technical Skills

- Version Control: Git (branching, merging, collaborative workflows).
- Programming Libraries: Python (NumPy, SciPy, scikit-learn, PyTorch), R (tidyverse, caret).
- Computational Tools: Jupyter Notebooks, LaTeX, data visualization packages (Matplotlib, ggplot2).
- Operating Systems: Linux (command line, shell scripting, environment configuration).

Languages

- Spanish: native proficiency.
- English: advanced reading, writing, and speaking skills.

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Research Experience

- Research Assistant in Actuarial Science (Konrad Lorenz University, Jan 2026 – Dec 2026): Currently working with six months of experience supporting actuarial research projects, including statistical modeling, risk analysis, and applications of probability and data science in financial and insurance contexts.
- Summer Research Program (Delfín Organization, Jun 2026 – Aug 2026): Three-month program resulting in a research article on clustering methods (K-means, DBSCAN) applied to ICFES Saber 11 datasets.
- Topological Data Analysis Project (2026 – ongoing): Research focused on the application of Topological Data Analysis (TDA) combined with statistical methods to real datasets from Colombia. Work includes the use of persistence diagrams, homology dimensions, Euler characteristic curves, and persistence landscapes to study structural properties of data. On the computational side, methods such as parallel programming, vectorization, and comparative approaches are employed to optimize performance.

Conferences and Presentations

- Speaker at 'Congreso Internacional del 31° Verano de la Investigación Científica y Tecnológica del Pacífico, realizado del 26 al 29 de agosto de 2026 Nuevo Nayarit, México' with the presentation "Clustering methods applied to Colombian educational datasets (ICFES Saber 11)." ID: 16131